

Supplementary material

Table S1: Supplementary Table S1. Complete paired 5 km block-RMSE inference for all 72 predefined Rolling-Origin contrasts. Δ is left minus right: a positive value means lower block RMSE for the longer right-hand configuration, whereas a negative value means higher block RMSE. Holm adjustment is within the corresponding three-contrast sensor-AOI-target family. CI = 95% confidence interval for mean Δ ; “not supported” means not statistically supported after the relevant Holm adjustment.

Sensor	AOI	Year	Contrast	n	Mean Δ	SD	SE	t	df	Raw p	Holm p	95% CI	Direction
Landsat 8/9	AOI-00	2024	H1-H2	39	0.000637	0.002087	0.000334	1.906	38	0.064240	0.064240	[-3.96e-05, 0.00131]	not supported
Landsat 8/9	AOI-00	2024	H1-H3	39	0.001031	0.001881	0.000301	3.424	38	0.001493	0.003762	[0.00042, 0.00164]	longer-window lower error
Landsat 8/9	AOI-00	2024	H2-H3	39	0.000395	0.000707	0.000113	3.486	38	0.001254	0.003762	[0.00017, 0.00062]	longer-window lower error
Landsat 8/9	AOI-00	2025	H1-H2	39	0.003216	0.005998	0.000960	3.348	38	0.001845	0.005536	[0.00127, 0.00516]	longer-window lower error
Landsat 8/9	AOI-00	2025	H1-H3	39	0.003360	0.007081	0.001134	2.963	38	0.005227	0.010455	[0.00106, 0.00566]	longer-window lower error
Landsat 8/9	AOI-00	2025	H2-H3	39	0.000145	0.001254	0.000201	0.720	38	0.475969	0.475969	[-0.00026, 0.00055]	not supported
Landsat 8/9	AOI-01	2024	H1-H2	35	3.51e-05	7.52e-05	1.27e-05	2.760	34	0.009237	0.009237	[9.26e-06, 6.09e-05]	longer-window lower error
Landsat 8/9	AOI-01	2024	H1-H3	35	5.08e-05	9.61e-05	1.62e-05	3.130	34	0.003580	0.007161	[1.78e-05, 8.39e-05]	longer-window lower error
Landsat 8/9	AOI-01	2024	H2-H3	35	1.57e-05	2.15e-05	3.63e-06	4.339	34	0.000121	0.000364	[8.37e-06, 2.31e-05]	longer-window lower error
Landsat 8/9	AOI-01	2025	H1-H2	35	8.97e-06	1.95e-05	3.30e-06	2.718	34	0.010260	0.030780	[2.26e-06, 1.57e-05]	longer-window lower error
Landsat 8/9	AOI-01	2025	H1-H3	35	2.07e-05	5.28e-05	8.93e-06	2.324	34	0.026212	0.052424	[2.61e-06, 3.89e-05]	not supported
Landsat 8/9	AOI-01	2025	H2-H3	35	1.18e-05	7.06e-05	1.19e-05	0.987	34	0.330787	0.330787	[-1.25e-05, 3.60e-05]	not supported
Landsat 8/9	AOI-02	2024	H1-H2	30	-0.000221	0.008015	0.001463	-0.151	29	0.881180	1.000000	[-0.00321, 0.00277]	not supported
Landsat 8/9	AOI-02	2024	H1-H3	30	-0.001147	0.010493	0.001916	-0.599	29	0.553981	1.000000	[-0.00507, 0.00277]	not supported
Landsat 8/9	AOI-02	2024	H2-H3	30	-0.000926	0.002556	0.000467	-1.985	29	0.056693	0.170078	[-0.00188, 2.82e-05]	not supported
Landsat 8/9	AOI-02	2025	H1-H2	36	-0.004124	0.003057	0.000510	-8.094	35	1.57e-09	4.71e-09	[-0.00516, -0.00309]	longer-window higher error
Landsat 8/9	AOI-02	2025	H1-H3	36	0.000898	0.001496	0.000249	3.600	35	0.000975	0.000975	[0.00039, 0.00140]	longer-window lower error
Landsat 8/9	AOI-02	2025	H2-H3	36	0.005022	0.004252	0.000709	7.087	35	2.95e-08	5.91e-08	[0.00358, 0.00646]	longer-window lower error
Landsat 8/9	AOI-03	2024	H1-H2	35	0.006086	0.004417	0.000747	8.151	34	1.66e-09	3.32e-09	[0.00457, 0.00760]	longer-window lower error

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Table S1 continued.

Sensor	AOI	Year	Contrast	n	Mean Δ	SD	SE	t	df	Raw p	Holm p	95% CI	Direction
Landsat 8/9	AOI-03	2024	H1-H3	35	0.005490	0.003640	0.000615	8.922	34	1.99e-10	5.97e-10	[0.00424, 0.00674]	longer-window error lower
Landsat 8/9	AOI-03	2024	H2-H3	35	-0.000596	0.000800	0.000135	-4.411	34	9.84e-05	9.84e-05	[-0.00087, -0.00032]	longer-window error higher error
Landsat 8/9	AOI-03	2025	H1-H2	35	-0.002232	0.002947	0.000498	-4.481	34	8.00e-05	0.000160	[-0.00324, -0.00122]	longer-window error higher error
Landsat 8/9	AOI-03	2025	H1-H3	35	-0.000489	0.000871	0.000147	-3.321	34	0.002154	0.002154	[-0.00079, -0.00019]	longer-window error higher error
Landsat 8/9	AOI-03	2025	H2-H3	35	0.001743	0.002207	0.000373	4.671	34	4.56e-05	0.000137	[0.00098, 0.00250]	longer-window error lower error
MODIS	AOI-00	2024	H1-H2	39	0.000287	0.001238	0.000198	1.445	38	0.156544	0.156544	[-0.00011, 0.00069]	not supported
MODIS	AOI-00	2024	H1-H3	39	0.001085	0.002169	0.000347	3.124	38	0.003411	0.006823	[0.00038, 0.00179]	longer-window error lower error
MODIS	AOI-00	2024	H2-H3	39	0.000798	0.001296	0.000207	3.848	38	0.000442	0.001327	[0.00038, 0.00122]	longer-window error lower error
MODIS	AOI-00	2025	H1-H2	39	0.002263	0.004622	0.000740	3.058	38	0.004065	0.012196	[0.00077, 0.00376]	longer-window error lower error
MODIS	AOI-00	2025	H1-H3	39	0.002513	0.005772	0.000924	2.719	38	0.009821	0.019642	[0.00064, 0.00438]	longer-window error lower error
MODIS	AOI-00	2025	H2-H3	39	0.000250	0.001191	0.000191	1.309	38	0.198412	0.198412	[-0.00014, 0.00064]	not supported
MODIS	AOI-01	2024	H1-H2	35	-2.18e-05	3.08e-05	5.20e-06	-4.186	34	0.000189	0.000189	[-3.23e-05, -1.12e-05]	longer-window error higher error
MODIS	AOI-01	2024	H1-H3	35	1.52e-05	1.10e-05	1.86e-06	8.145	34	1.69e-09	5.06e-09	[1.14e-05, 1.89e-05]	longer-window error lower error
MODIS	AOI-01	2024	H2-H3	35	3.69e-05	4.06e-05	6.87e-06	5.376	34	5.57e-06	1.11e-05	[2.30e-05, 5.09e-05]	longer-window error lower error
MODIS	AOI-01	2025	H1-H2	35	2.99e-05	0.000121	2.05e-05	1.460	34	0.153400	0.460201	[-1.17e-05, 7.15e-05]	not supported
MODIS	AOI-01	2025	H1-H3	35	3.70e-05	0.000185	3.12e-05	1.186	34	0.243679	0.487357	[-2.64e-05, 0.00010]	not supported
MODIS	AOI-01	2025	H2-H3	35	7.10e-06	6.35e-05	1.07e-05	0.662	34	0.512323	0.512323	[-1.47e-05, 2.89e-05]	not supported
MODIS	AOI-02	2024	H1-H2	38	0.004723	0.004444	0.000721	6.551	37	1.13e-07	3.39e-07	[0.00326, 0.00618]	longer-window error lower error
MODIS	AOI-02	2024	H1-H3	38	0.004904	0.004929	0.000800	6.133	37	4.15e-07	8.30e-07	[0.00328, 0.00652]	longer-window error lower error
MODIS	AOI-02	2024	H2-H3	38	0.000181	0.000674	0.000109	1.657	37	0.106004	0.106004	[-4.04e-05, 0.00040]	not supported
MODIS	AOI-02	2025	H1-H2	38	0.003613	0.001638	0.000266	13.602	37	5.70e-16	1.14e-15	[0.00308, 0.00415]	longer-window error lower error

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Table S1 continued.

Sensor	AOI	Year	Contrast	<i>n</i>	Mean Δ	SD	SE	<i>t</i>	df	Raw <i>p</i>	Holm <i>p</i>	95% CI	Direction
MODIS	AOI-02	2025	H1-H3	38	0.002434	0.001087	0.000176	13.799	37	3.65e-16	1.10e-15	[0.00208, 0.00279]	longer-window lower error
MODIS	AOI-02	2025	H2-H3	38	-0.001179	0.000617	0.000100	-11.773	37	4.47e-14	4.47e-14	[-0.00138, -0.00098]	longer-window higher error
MODIS	AOI-03	2024	H1-H2	35	0.003257	0.000843	0.000143	22.852	34	3.25e-22	9.75e-22	[0.00297, 0.00355]	longer-window lower error
MODIS	AOI-03	2024	H1-H3	35	0.007654	0.002310	0.000390	19.607	34	4.15e-20	8.30e-20	[0.00686, 0.00845]	longer-window lower error
MODIS	AOI-03	2024	H2-H3	35	0.004397	0.001494	0.000253	17.408	34	1.66e-18	1.66e-18	[0.00388, 0.00491]	longer-window lower error
MODIS	AOI-03	2025	H1-H2	35	-0.001038	0.006592	0.001114	-0.932	34	0.357955	0.575259	[-0.00330, 0.00123]	not supported
MODIS	AOI-03	2025	H1-H3	35	-0.001318	0.007219	0.001220	-1.080	34	0.287629	0.575259	[-0.00380, 0.00116]	not supported
MODIS	AOI-03	2025	H2-H3	35	-0.000280	0.000638	0.000108	-2.593	34	0.013916	0.041749	[-0.00050, -6.06e-05]	longer-window higher error
Sentinel-2	AOI-00	2024	H1-H2	39	0.000296	0.000737	0.000118	2.509	38	0.016509	0.049528	[5.72e-05, 0.00054]	longer-window lower error
Sentinel-2	AOI-00	2024	H1-H3	39	-0.000169	0.005695	0.000912	-0.186	38	0.853693	1.000000	[-0.00202, 0.00168]	not supported
Sentinel-2	AOI-00	2024	H2-H3	39	-0.000466	0.005125	0.000821	-0.567	38	0.573894	1.000000	[-0.00213, 0.00120]	not supported
Sentinel-2	AOI-00	2025	H1-H2	39	0.000777	0.004294	0.000688	1.130	38	0.265589	0.796768	[-0.00062, 0.00217]	not supported
Sentinel-2	AOI-00	2025	H1-H3	39	0.000835	0.004666	0.000747	1.118	38	0.270750	0.796768	[-0.00068, 0.00235]	not supported
Sentinel-2	AOI-00	2025	H2-H3	39	5.81e-05	0.000376	6.02e-05	0.964	38	0.340976	0.796768	[-6.39e-05, 0.00018]	not supported
Sentinel-2	AOI-01	2024	H1-H2	35	-4.32e-05	3.11e-05	5.26e-06	-8.211	34	1.40e-09	2.80e-09	[-5.39e-05, -3.25e-05]	longer-window higher error
Sentinel-2	AOI-01	2024	H1-H3	35	-6.55e-06	1.95e-05	3.29e-06	-1.993	34	0.054367	0.054367	[-1.32e-05, 1.30e-07]	not supported
Sentinel-2	AOI-01	2024	H2-H3	35	3.67e-05	2.09e-05	3.53e-06	10.372	34	4.54e-12	1.36e-11	[2.95e-05, 4.38e-05]	longer-window lower error
Sentinel-2	AOI-01	2025	H1-H2	35	1.62e-05	3.11e-05	5.26e-06	3.075	34	0.004132	0.012396	[5.49e-06, 2.69e-05]	longer-window lower error
Sentinel-2	AOI-01	2025	H1-H3	35	1.92e-05	4.72e-05	7.99e-06	2.403	34	0.021873	0.043745	[2.96e-06, 3.54e-05]	longer-window lower error
Sentinel-2	AOI-01	2025	H2-H3	35	3.00e-06	2.16e-05	3.66e-06	0.821	34	0.417265	0.417265	[-4.43e-06, 1.04e-05]	not supported
Sentinel-2	AOI-02	2024	H1-H2	38	-0.000362	0.008130	0.001319	-0.275	37	0.785099	0.785099	[-0.00303, 0.00231]	not supported
Sentinel-2	AOI-02	2024	H1-H3	38	0.000625	0.002913	0.000473	1.322	37	0.194310	0.582930	[-0.00033, 0.00158]	not supported
Sentinel-2	AOI-02	2024	H2-H3	38	0.000987	0.005308	0.000861	1.146	37	0.259059	0.582930	[-0.00076, 0.00273]	not supported
Sentinel-2	AOI-02	2025	H1-H2	38	-0.000400	0.000933	0.000151	-2.642	37	0.012008	0.036024	[-0.00071, -9.32e-05]	longer-window higher error
Sentinel-2	AOI-02	2025	H1-H3	38	0.000366	0.003641	0.000591	0.620	37	0.539045	0.598943	[-0.00083, 0.00156]	not supported
Sentinel-2	AOI-02	2025	H2-H3	38	0.000766	0.004488	0.000728	1.052	37	0.299471	0.598943	[-0.00071, 0.00224]	not supported
Sentinel-2	AOI-03	2024	H1-H2	35	-6.15e-05	0.000473	8.00e-05	-0.769	34	0.447487	0.570595	[-0.00022, 0.00010]	not supported

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Table S1 continued.

Sensor	AOI	Year	Contrast	n	Mean Δ	SD	SE	t	df	Raw p	Holm p	95% CI	Direction
Sentinel-2	AOI-03	2024	H1-H3	35	-0.000304	0.001538	0.000260	-1.171	34	0.249731	0.570595	[-0.00083, 0.00022]	not supported
Sentinel-2	AOI-03	2024	H2-H3	35	-0.000243	0.001075	0.000182	-1.337	34	0.190198	0.570595	[-0.00061, 0.00013]	not supported
Sentinel-2	AOI-03	2025	H1-H2	35	0.000302	0.001170	0.000198	1.529	34	0.135565	0.186165	[-9.96e-05, 0.00070]	not supported
Sentinel-2	AOI-03	2025	H1-H3	35	0.000373	0.001276	0.000216	1.728	34	0.093083	0.186165	[-6.56e-05, 0.00081]	not supported
Sentinel-2	AOI-03	2025	H2-H3	35	7.03e-05	0.000185	3.12e-05	2.252	34	0.030920	0.092761	[6.85e-06, 0.00013]	not supported

Table S2: LOYO diagnostic summaries across completed held-out constituent-year fits. Each row summarizes all multi-year windows and held-out years for that sensor-AOI; entries are min / median / max. LOYO is an internal temporal diagnostic and was not used for window ranking.

Sensor	AOI	Fits	Held-out n	RMSE	MAE	Bias	R^2
Landsat 8/9	AOI-00	7	15212 / 16142 / 16682	0.0481 / 0.0510 / 0.0531	0.0384 / 0.0413 / 0.0431	-0.0223 / -0.0028 / 0.0224	0.6707 / 0.7444 / 0.7633
Landsat 8/9	AOI-01	7	14773 / 15451 / 15456	0.0013 / 0.0013 / 0.0014	0.0004 / 0.0005 / 0.0006	0.0002 / 0.0002 / 0.0004	0.4357 / 0.4798 / 0.4890
Landsat 8/9	AOI-02	7	554 / 2134 / 5990	0.0536 / 0.0687 / 0.0751	0.0411 / 0.0549 / 0.0606	-0.0290 / -0.0085 / 0.0269	0.7698 / 0.8018 / 0.8393
Landsat 8/9	AOI-03	7	8400 / 12075 / 12706	0.0329 / 0.0395 / 0.0456	0.0233 / 0.0315 / 0.0339	-0.0239 / 0.0033 / 0.0238	0.8265 / 0.8686 / 0.9020
MODIS	AOI-00	7	16719 / 16719 / 16719	0.0384 / 0.0433 / 0.0457	0.0299 / 0.0342 / 0.0374	-0.0182 / -0.0038 / 0.0181	0.7569 / 0.8164 / 0.8515
MODIS	AOI-01	7	15456 / 15456 / 15456	0.0014 / 0.0015 / 0.0017	0.0003 / 0.0005 / 0.0006	7.65e-05 / 0.0002 / 0.0003	0.2000 / 0.2653 / 0.3076
MODIS	AOI-02	7	7512 / 12912 / 13402	0.0393 / 0.0485 / 0.0580	0.0302 / 0.0383 / 0.0447	-0.0237 / -0.0075 / 0.0210	0.8589 / 0.9156 / 0.9420
MODIS	AOI-03	7	14011 / 14562 / 14703	0.0238 / 0.0324 / 0.0447	0.0166 / 0.0236 / 0.0331	-0.0259 / -0.0039 / 0.0288	0.8147 / 0.8975 / 0.9332
Sentinel-2	AOI-00	7	12687 / 15086 / 16713	0.0452 / 0.0610 / 0.0624	0.0362 / 0.0486 / 0.0493	-0.0057 / -0.0019 / 0.0036	0.5464 / 0.6347 / 0.7931
Sentinel-2	AOI-01	7	15310 / 15456 / 15456	0.0013 / 0.0014 / 0.0016	0.0004 / 0.0005 / 0.0006	0.0001 / 0.0002 / 0.0004	0.3133 / 0.3771 / 0.4352
Sentinel-2	AOI-02	7	6328 / 14195 / 14595	0.0574 / 0.0614 / 0.0806	0.0447 / 0.0480 / 0.0610	-0.0331 / -0.0104 / 0.0330	0.7887 / 0.8650 / 0.8823
Sentinel-2	AOI-03	7	13060 / 14910 / 14925	0.0354 / 0.0360 / 0.0419	0.0242 / 0.0248 / 0.0292	-0.0041 / 0.0009 / 0.0042	0.8147 / 0.8574 / 0.8815

Table S3: Pre-refit held-out historical spatial-reserve diagnostic summaries. Each row summarizes six historical windows for that sensor–AOI; entries are min / median / max. The reserve was not used for tuning, window selection, or inference and was recombined before the prescribed full historical fit.

Sensor	AOI	Fits	Reserve n	RMSE	MAE	Bias	R^2
Landsat 8/9	AOI-00	6	3376 / 5147 / 10294	0.0455 / 0.0472 / 0.0489	0.0369 / 0.0381 / 0.0406	0.0054 / 0.0062 / 0.0073	0.7220 / 0.7505 / 0.7686
Landsat 8/9	AOI-01	6	4376 / 6610 / 13219	0.0008 / 0.0008 / 0.0008	0.0005 / 0.0006 / 0.0006	0.0005 / 0.0005 / 0.0005	-1.4932 / -1.0736 / -0.9559
Landsat 8/9	AOI-02	6	276 / 1348 / 3274	0.0519 / 0.0596 / 0.0696	0.0387 / 0.0445 / 0.0535	0.0019 / 0.0039 / 0.0162	0.8207 / 0.8530 / 0.9080
Landsat 8/9	AOI-03	6	2778 / 4889 / 9778	0.0238 / 0.0299 / 0.0340	0.0173 / 0.0220 / 0.0254	0.0039 / 0.0051 / 0.0083	0.8575 / 0.8673 / 0.8931
MODIS	AOI-00	6	3537 / 5306 / 10611	0.0358 / 0.0407 / 0.0432	0.0294 / 0.0330 / 0.0349	0.0014 / 0.0034 / 0.0058	0.7906 / 0.8232 / 0.8633
MODIS	AOI-01	6	4422 / 6633 / 13266	0.0006 / 0.0007 / 0.0007	0.0004 / 0.0005 / 0.0005	0.0003 / 0.0004 / 0.0005	-0.8175 / -0.4794 / -0.2556
MODIS	AOI-02	6	2041 / 5652 / 11282	0.0425 / 0.0481 / 0.0567	0.0315 / 0.0360 / 0.0428	0.0025 / 0.0043 / 0.0100	0.8817 / 0.9292 / 0.9437
MODIS	AOI-03	6	3956 / 6122 / 12244	0.0222 / 0.0265 / 0.0289	0.0152 / 0.0187 / 0.0204	-0.0016 / 0.0026 / 0.0033	0.8882 / 0.9022 / 0.9110
Sentinel-2	AOI-00	6	3132 / 5099 / 9819	0.0470 / 0.0535 / 0.0585	0.0382 / 0.0428 / 0.0468	0.0021 / 0.0079 / 0.0140	0.5722 / 0.6741 / 0.7624
Sentinel-2	AOI-01	6	4419 / 6632 / 13263	0.0006 / 0.0007 / 0.0007	0.0004 / 0.0004 / 0.0005	0.0003 / 0.0003 / 0.0004	-0.8930 / -0.5679 / -0.2371
Sentinel-2	AOI-02	6	1258 / 4837 / 8855	0.0530 / 0.0629 / 0.0842	0.0411 / 0.0476 / 0.0660	0.0035 / 0.0045 / 0.0077	0.8195 / 0.8831 / 0.9124
Sentinel-2	AOI-03	6	3671 / 6122 / 12245	0.0297 / 0.0336 / 0.0378	0.0215 / 0.0247 / 0.0285	0.0054 / 0.0075 / 0.0093	0.7984 / 0.8385 / 0.8632

Table S4: Full four-endpoint DPM sensitivity matrix. Raw predictions are calculated before clipping to $[0, 1]$.

Sensor	AOI	Endpoint	$NDVI_{low}$	$NDVI_{high}$	Spread	n	RMSE	Bias	Clip.	$\Delta RMSE$ clip.
Landsat 8/9	AOI-00	P1/P99	0.2226	0.7870	0.5644	20,251	0.2278	0.2052	0.020	0.0011
Landsat 8/9	AOI-00	P2/P98	0.3053	0.7761	0.4709	20,251	0.2206	0.1851	0.040	0.0060
Landsat 8/9	AOI-00	P5/P95	0.4636	0.7611	0.2974	20,251	0.2061	0.1063	0.100	0.0631
Landsat 8/9	AOI-00	P10/P90	0.5628	0.7458	0.1829	20,251	0.2452	0.0170	0.200	0.2217
Landsat 8/9	AOI-01	P1/P99	0.0348	0.0784	0.0436	14,738	0.4860	0.4497	0.020	0.0139
Landsat 8/9	AOI-01	P2/P98	0.0384	0.0738	0.0354	14,738	0.5020	0.4519	0.040	0.0242
Landsat 8/9	AOI-01	P5/P95	0.0419	0.0685	0.0266	14,738	0.5394	0.4677	0.100	0.0518
Landsat 8/9	AOI-01	P10/P90	0.0446	0.0643	0.0198	14,738	0.5843	0.4891	0.200	0.1089
Landsat 8/9	AOI-02	P1/P99	0.1478	0.7926	0.6447	7,600	0.2111	0.1867	0.020	0.0004
Landsat 8/9	AOI-02	P2/P98	0.1720	0.7798	0.6077	7,600	0.2172	0.1883	0.040	0.0011
Landsat 8/9	AOI-02	P5/P95	0.2501	0.7532	0.5031	7,600	0.2251	0.1780	0.100	0.0081
Landsat 8/9	AOI-02	P10/P90	0.3732	0.7292	0.3560	7,600	0.2291	0.1290	0.200	0.0594
Landsat 8/9	AOI-03	P1/P99	0.0831	0.5880	0.5050	14,402	0.2097	0.1722	0.020	0.0014
Landsat 8/9	AOI-03	P2/P98	0.0902	0.5520	0.4618	14,402	0.2283	0.1842	0.040	0.0045
Landsat 8/9	AOI-03	P5/P95	0.1039	0.4669	0.3630	14,402	0.2842	0.2252	0.100	0.0249
Landsat 8/9	AOI-03	P10/P90	0.1178	0.3754	0.2576	14,402	0.3719	0.2953	0.200	0.0881
MODIS	AOI-00	P1/P99	0.3066	0.8064	0.4998	20,256	0.2076	0.1861	0.020	0.0016
MODIS	AOI-00	P2/P98	0.3920	0.7970	0.4050	20,256	0.1948	0.1538	0.040	0.0085
MODIS	AOI-00	P5/P95	0.5189	0.7829	0.2640	20,256	0.1954	0.0711	0.100	0.0631
MODIS	AOI-00	P10/P90	0.5891	0.7697	0.1806	20,256	0.2436	-0.0014	0.200	0.1692
MODIS	AOI-01	P1/P99	0.0432	0.0839	0.0406	19,878	0.4303	0.3816	0.020	0.0103
MODIS	AOI-01	P2/P98	0.0452	0.0791	0.0339	19,878	0.4602	0.3977	0.040	0.0185
MODIS	AOI-01	P5/P95	0.0472	0.0736	0.0264	19,878	0.5135	0.4314	0.100	0.0407
MODIS	AOI-01	P10/P90	0.0488	0.0696	0.0208	19,878	0.5627	0.4630	0.200	0.0817
MODIS	AOI-02	P1/P99	0.1710	0.8073	0.6363	16,568	0.2040	0.1833	0.020	0.0004
MODIS	AOI-02	P2/P98	0.1903	0.7941	0.6038	16,568	0.2121	0.1875	0.040	0.0010
MODIS	AOI-02	P5/P95	0.2493	0.7738	0.5244	16,568	0.2177	0.1796	0.100	0.0053
MODIS	AOI-02	P10/P90	0.3547	0.7497	0.3950	16,568	0.2216	0.1465	0.200	0.0386
MODIS	AOI-03	P1/P99	0.1016	0.5614	0.4598	17,860	0.2193	0.1895	0.020	0.0014
MODIS	AOI-03	P2/P98	0.1093	0.5248	0.4155	17,860	0.2409	0.2041	0.040	0.0047
MODIS	AOI-03	P5/P95	0.1214	0.4496	0.3283	17,860	0.3003	0.2500	0.100	0.0235
MODIS	AOI-03	P10/P90	0.1369	0.3699	0.2330	17,860	0.3908	0.3208	0.200	0.0806
Sentinel-2	AOI-00	P1/P99	0.1479	0.7869	0.6390	19,233	0.1973	0.1556	0.020	0.0014
Sentinel-2	AOI-00	P2/P98	0.2307	0.7759	0.5452	19,233	0.1936	0.1300	0.040	0.0060
Sentinel-2	AOI-00	P5/P95	0.3780	0.7561	0.3781	19,233	0.2084	0.0592	0.100	0.0427
Sentinel-2	AOI-00	P10/P90	0.4700	0.7340	0.2640	19,233	0.2585	-0.0008	0.200	0.1262
Sentinel-2	AOI-01	P1/P99	0.0120	0.0523	0.0403	19,878	0.4215	0.3780	0.020	0.0652
Sentinel-2	AOI-01	P2/P98	0.0142	0.0464	0.0322	19,878	0.4622	0.4042	0.040	0.0962
Sentinel-2	AOI-01	P5/P95	0.0165	0.0405	0.0240	19,878	0.5202	0.4398	0.100	0.1605
Sentinel-2	AOI-01	P10/P90	0.0183	0.0368	0.0185	19,878	0.5686	0.4676	0.200	0.2545
Sentinel-2	AOI-02	P1/P99	0.0708	0.8029	0.7322	18,534	0.1800	0.1429	0.020	0.0004
Sentinel-2	AOI-02	P2/P98	0.0881	0.7865	0.6984	18,534	0.1901	0.1483	0.040	0.0011
Sentinel-2	AOI-02	P5/P95	0.1232	0.7575	0.6344	18,534	0.2087	0.1569	0.100	0.0045
Sentinel-2	AOI-02	P10/P90	0.2148	0.7272	0.5124	18,534	0.2222	0.1393	0.200	0.0247
Sentinel-2	AOI-03	P1/P99	0.0428	0.5620	0.5192	18,857	0.2075	0.1693	0.020	0.0024
Sentinel-2	AOI-03	P2/P98	0.0510	0.5105	0.4595	18,857	0.2357	0.1892	0.040	0.0070
Sentinel-2	AOI-03	P5/P95	0.0635	0.4188	0.3553	18,857	0.3029	0.2405	0.100	0.0290
Sentinel-2	AOI-03	P10/P90	0.0762	0.3436	0.2675	18,857	0.3781	0.2988	0.200	0.0791

Table S5: Clipping diagnostics for all DPM endpoint configurations and OLS histories: rates and RMSE.

Model	Sensor	AOI/configuration	n	Raw < 0	Raw > 1	Clipped	RMSE raw	RMSE clipped	ΔRMSE
DPM	Landsat 8/9	AOI-00; P1/P99	20,251	0.010	0.010	0.020	0.2289	0.2278	0.0011
DPM	Landsat 8/9	AOI-00; P10/P90	20,251	0.100	0.100	0.200	0.4669	0.2452	0.2217
DPM	Landsat 8/9	AOI-00; P2/P98	20,251	0.020	0.020	0.040	0.2267	0.2206	0.0060
DPM	Landsat 8/9	AOI-00; P5/P95	20,251	0.050	0.050	0.100	0.2692	0.2061	0.0631
DPM	Landsat 8/9	AOI-01; P1/P99	14,738	0.010	0.010	0.020	0.4999	0.4860	0.0139
DPM	Landsat 8/9	AOI-01; P10/P90	14,738	0.100	0.100	0.200	0.6932	0.5843	0.1089
DPM	Landsat 8/9	AOI-01; P2/P98	14,738	0.020	0.020	0.040	0.5262	0.5020	0.0242
DPM	Landsat 8/9	AOI-01; P5/P95	14,738	0.050	0.050	0.100	0.5912	0.5394	0.0518
DPM	Landsat 8/9	AOI-02; P1/P99	7,600	0.010	0.010	0.020	0.2115	0.2111	0.0004
DPM	Landsat 8/9	AOI-02; P10/P90	7,600	0.100	0.100	0.200	0.2886	0.2291	0.0594
DPM	Landsat 8/9	AOI-02; P2/P98	7,600	0.020	0.020	0.040	0.2183	0.2172	0.0011
DPM	Landsat 8/9	AOI-02; P5/P95	7,600	0.050	0.050	0.100	0.2332	0.2251	0.0081
DPM	Landsat 8/9	AOI-03; P1/P99	14,402	0.010	0.010	0.020	0.2111	0.2097	0.0014
DPM	Landsat 8/9	AOI-03; P10/P90	14,402	0.100	0.100	0.200	0.4599	0.3719	0.0881
DPM	Landsat 8/9	AOI-03; P2/P98	14,402	0.020	0.020	0.040	0.2328	0.2283	0.0045
DPM	Landsat 8/9	AOI-03; P5/P95	14,402	0.050	0.050	0.100	0.3090	0.2842	0.0249
DPM	MODIS	AOI-00; P1/P99	20,256	0.010	0.010	0.020	0.2092	0.2076	0.0016
DPM	MODIS	AOI-00; P10/P90	20,256	0.100	0.100	0.200	0.4129	0.2436	0.1692
DPM	MODIS	AOI-00; P2/P98	20,256	0.020	0.020	0.040	0.2033	0.1948	0.0085
DPM	MODIS	AOI-00; P5/P95	20,256	0.050	0.050	0.100	0.2585	0.1954	0.0631
DPM	MODIS	AOI-01; P1/P99	19,878	0.010	0.010	0.020	0.4407	0.4303	0.0103
DPM	MODIS	AOI-01; P10/P90	19,878	0.100	0.100	0.200	0.6444	0.5627	0.0817
DPM	MODIS	AOI-01; P2/P98	19,878	0.020	0.020	0.040	0.4787	0.4602	0.0185
DPM	MODIS	AOI-01; P5/P95	19,878	0.050	0.050	0.100	0.5542	0.5135	0.0407
DPM	MODIS	AOI-02; P1/P99	16,568	0.010	0.010	0.020	0.2043	0.2040	0.0004
DPM	MODIS	AOI-02; P10/P90	16,568	0.100	0.100	0.200	0.2602	0.2216	0.0386
DPM	MODIS	AOI-02; P2/P98	16,568	0.020	0.020	0.040	0.2130	0.2121	0.0010
DPM	MODIS	AOI-02; P5/P95	16,568	0.050	0.050	0.100	0.2230	0.2177	0.0053
DPM	MODIS	AOI-03; P1/P99	17,860	0.010	0.010	0.020	0.2208	0.2193	0.0014
DPM	MODIS	AOI-03; P10/P90	17,860	0.100	0.100	0.200	0.4714	0.3908	0.0806
DPM	MODIS	AOI-03; P2/P98	17,860	0.020	0.020	0.040	0.2456	0.2409	0.0047
DPM	MODIS	AOI-03; P5/P95	17,860	0.050	0.050	0.100	0.3238	0.3003	0.0235
DPM	Sentinel-2	AOI-00; P1/P99	19,233	0.010	0.010	0.020	0.1986	0.1973	0.0014
DPM	Sentinel-2	AOI-00; P10/P90	19,233	0.100	0.100	0.200	0.3848	0.2585	0.1262
DPM	Sentinel-2	AOI-00; P2/P98	19,233	0.020	0.020	0.040	0.1996	0.1936	0.0060
DPM	Sentinel-2	AOI-00; P5/P95	19,233	0.050	0.050	0.100	0.2511	0.2084	0.0427
DPM	Sentinel-2	AOI-01; P1/P99	19,878	0.010	0.010	0.020	0.4867	0.4215	0.0652
DPM	Sentinel-2	AOI-01; P10/P90	19,878	0.100	0.100	0.200	0.8232	0.5686	0.2545
DPM	Sentinel-2	AOI-01; P2/P98	19,878	0.020	0.020	0.040	0.5584	0.4622	0.0962
DPM	Sentinel-2	AOI-01; P5/P95	19,878	0.050	0.050	0.100	0.6806	0.5202	0.1605
DPM	Sentinel-2	AOI-02; P1/P99	18,534	0.010	0.010	0.020	0.1804	0.1800	0.0004
DPM	Sentinel-2	AOI-02; P10/P90	18,534	0.100	0.100	0.200	0.2469	0.2222	0.0247
DPM	Sentinel-2	AOI-02; P2/P98	18,534	0.020	0.020	0.040	0.1911	0.1901	0.0011
DPM	Sentinel-2	AOI-02; P5/P95	18,534	0.050	0.050	0.100	0.2132	0.2087	0.0045
DPM	Sentinel-2	AOI-03; P1/P99	18,557	0.010	0.010	0.020	0.2098	0.2075	0.0024
DPM	Sentinel-2	AOI-03; P10/P90	18,557	0.100	0.100	0.200	0.4572	0.3781	0.0791
DPM	Sentinel-2	AOI-03; P2/P98	18,557	0.020	0.020	0.040	0.2427	0.2357	0.0070
DPM	Sentinel-2	AOI-03; P5/P95	18,557	0.050	0.050	0.100	0.3319	0.3029	0.0290
OLS	Landsat 8/9	AOI-00; OLS (W2022)	20,251	0.000	0.000	0.000	0.0486	0.0486	0.0000
OLS	Landsat 8/9	AOI-00; OLS (W2022_2023)	20,251	0.000	0.000	0.000	0.0480	0.0480	0.0000
OLS	Landsat 8/9	AOI-00; OLS (W2022_2024)	20,251	0.000	0.000	0.000	0.0494	0.0494	0.0000
OLS	Landsat 8/9	AOI-00; OLS (W2023)	20,251	0.000	0.000	0.000	0.0476	0.0476	0.0000
OLS	Landsat 8/9	AOI-00; OLS (W2023_2024)	20,251	0.000	0.000	0.000	0.0499	0.0499	0.0000
OLS	Landsat 8/9	AOI-00; OLS (W2024)	20,251	0.000	0.000	0.000	0.0546	0.0546	0.0000
OLS	Landsat 8/9	AOI-01; OLS (W2022)	14,738	0.434	0.000	0.434	0.0015	0.0014	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2022_2023)	14,738	0.402	0.000	0.402	0.0015	0.0014	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2022_2024)	14,738	0.382	0.000	0.382	0.0015	0.0014	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2023)	14,738	0.377	0.000	0.377	0.0015	0.0014	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2023_2024)	14,738	0.362	0.000	0.362	0.0015	0.0014	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2024)	14,738	0.347	0.000	0.347	0.0015	0.0014	0.0001
OLS	Landsat 8/9	AOI-02; OLS (W2022)	7,600	0.000	0.000	0.000	0.0564	0.0564	0.0000
OLS	Landsat 8/9	AOI-02; OLS (W2022_2023)	7,600	0.000	0.000	0.000	0.0571	0.0571	0.0000
OLS	Landsat 8/9	AOI-02; OLS (W2022_2024)	7,600	0.000	0.000	0.000	0.0574	0.0574	0.0000
OLS	Landsat 8/9	AOI-02; OLS (W2023)	7,600	0.000	0.000	0.000	0.0622	0.0622	0.0000
OLS	Landsat 8/9	AOI-02; OLS (W2023_2024)	7,600	0.000	0.000	0.000	0.0622	0.0622	0.0000
OLS	Landsat 8/9	AOI-02; OLS (W2024)	7,600	0.000	0.000	0.000	0.0583	0.0583	0.0000
OLS	Landsat 8/9	AOI-03; OLS (W2022)	14,402	0.016	0.000	0.016	0.0354	0.0353	0.0001
OLS	Landsat 8/9	AOI-03; OLS (W2022_2023)	14,402	0.012	0.000	0.012	0.0366	0.0365	0.0001
OLS	Landsat 8/9	AOI-03; OLS (W2022_2024)	14,402	0.010	0.000	0.010	0.0363	0.0362	0.0001
OLS	Landsat 8/9	AOI-03; OLS (W2023)	14,402	0.006	0.000	0.006	0.0424	0.0424	0.0000
OLS	Landsat 8/9	AOI-03; OLS (W2023_2024)	14,402	0.007	0.000	0.007	0.0378	0.0377	0.0001
OLS	Landsat 8/9	AOI-03; OLS (W2024)	14,402	0.009	0.000	0.009	0.0360	0.0360	0.0001
OLS	MODIS	AOI-00; OLS (W2022)	20,256	0.000	0.000	0.000	0.0389	0.0389	0.0000
OLS	MODIS	AOI-00; OLS (W2022_2023)	20,256	0.000	0.000	0.000	0.0389	0.0389	0.0000
OLS	MODIS	AOI-00; OLS (W2022_2024)	20,256	0.000	0.000	0.000	0.0395	0.0395	0.0000
OLS	MODIS	AOI-00; OLS (W2023)	20,256	0.000	0.000	0.000	0.0389	0.0389	0.0000
OLS	MODIS	AOI-00; OLS (W2023_2024)	20,256	0.000	0.000	0.000	0.0400	0.0400	0.0000
OLS	MODIS	AOI-00; OLS (W2024)	20,256	0.000	0.000	0.000	0.0432	0.0432	0.0000
OLS	MODIS	AOI-01; OLS (W2022)	19,878	0.361	0.000	0.361	0.0015	0.0014	0.0001
OLS	MODIS	AOI-01; OLS (W2022_2023)	19,878	0.375	0.000	0.375	0.0015	0.0014	0.0001
OLS	MODIS	AOI-01; OLS (W2022_2024)	19,878	0.373	0.000	0.373	0.0015	0.0015	0.0000
OLS	MODIS	AOI-01; OLS (W2023)	19,878	0.391	0.000	0.391	0.0015	0.0014	0.0001
OLS	MODIS	AOI-01; OLS (W2023_2024)	19,878	0.376	0.000	0.376	0.0015	0.0015	0.0000
OLS	MODIS	AOI-01; OLS (W2024)	19,878	0.351	0.000	0.351	0.0015	0.0015	0.0000
OLS	MODIS	AOI-02; OLS (W2022)	16,568	0.000	0.000	0.000	0.0542	0.0542	0.0000
OLS	MODIS	AOI-02; OLS (W2022_2023)	16,568	0.000	0.000	0.000	0.0511	0.0511	0.0000
OLS	MODIS	AOI-02; OLS (W2022_2024)	16,568	0.000	0.000	0.000	0.0524	0.0524	0.0000
OLS	MODIS	AOI-02; OLS (W2023)	16,568	0.004	0.000	0.004	0.0482	0.0482	0.0000
OLS	MODIS	AOI-02; OLS (W2023_2024)	16,568	0.000	0.000	0.000	0.0513	0.0513	0.0000
OLS	MODIS	AOI-02; OLS (W2024)	16,568	0.000	0.000	0.000	0.0547	0.0547	0.0000
OLS	MODIS	AOI-03; OLS (W2022)	17,860	0.013	0.000	0.013	0.0293	0.0292	0.0001
OLS	MODIS	AOI-03; OLS (W2022_2023)	17,860	0.012	0.000	0.012	0.0311	0.0310	0.0001
OLS	MODIS	AOI-03; OLS (W2022_2024)	17,860	0.013	0.000	0.013	0.0287	0.0286	0.0001
OLS	MODIS	AOI-03; OLS (W2023)	17,860	0.010	0.000	0.010	0.0334	0.0334	0.0001
OLS	MODIS	AOI-03; OLS (W2023_2024)	17,860	0.014	0.000	0.014	0.0285	0.0284	0.0001
OLS	MODIS	AOI-03; OLS (W2024)	17,860	0.031	0.000	0.031	0.0285	0.0283	0.0002
OLS	Sentinel-2	AOI-00; OLS (W2022)	19,233	0.000	0.000	0.000	0.0631	0.0631	0.0000
OLS	Sentinel-2	AOI-00; OLS (W2022_2023)	19,233	0.000	0.000	0.000	0.0632	0.0632	0.0000
OLS	Sentinel-2	AOI-00; OLS (W2022_2024)	19,233	0.000	0.000	0.000	0.0635	0.0635	0.0000
OLS	Sentinel-2	AOI-00; OLS (W2023)	19,233	0.000	0.000	0.000	0.0635	0.0635	0.0000
OLS	Sentinel-2	AOI-00; OLS (W2023_2024)	19,233	0.000	0.000	0.000	0.0637	0.0637	0.0000
OLS	Sentinel-2	AOI-00; OLS (W2024)	19,233	0.000	0.000	0.000	0.0658	0.0658	0.0000
OLS	Sentinel-2	AOI-01; OLS (W2022)	19,878	0.647	0.000	0.647	0.0015	0.0013	0.0002
OLS	Sentinel-2	AOI-01; OLS (W2022_2023)	19,878	0.714	0.000	0.714	0.0015	0.0013	0.0002
OLS	Sentinel-2	AOI-01; OLS (W2022_2024)	19,878	0.726	0.000	0.726	0.0015	0.0014	0.0002
OLS	Sentinel-2	AOI-01; OLS (W2023)	19,878	0.773	0.000	0.773	0.0016	0.0013	0.0002
OLS	Sentinel-2	AOI-01; OLS (W2023_2024)	19,878	0.761	0.000	0.761	0.0016	0.0014	0.0002
OLS	Sentinel-2	AOI-01; OLS (W2024)	19,878	0.745	0.000	0.745	0.0016	0.0014	0.0002

Model	Sensor	AOI/configuration	n	Raw < 0	Raw > 1	Clipped	RMSE raw	RMSE clipped	Δ RMSE
OLS	Sentinel-2	AOI-02; OLS (W2023)	18,534	0.000	0.000	0.000	0.0605	0.0605	0.0000
OLS	Sentinel-2	AOI-02; OLS (W2023_2024)	18,534	0.000	0.000	0.000	0.0589	0.0589	0.0000
OLS	Sentinel-2	AOI-02; OLS (W2024)	18,534	0.000	0.000	0.000	0.0586	0.0586	0.0000
OLS	Sentinel-2	AOI-03; OLS (W2022)	18,857	0.011	0.000	0.011	0.0396	0.0395	0.0001
OLS	Sentinel-2	AOI-03; OLS (W2022_2023)	18,857	0.011	0.000	0.011	0.0397	0.0396	0.0001
OLS	Sentinel-2	AOI-03; OLS (W2022_2024)	18,857	0.010	0.000	0.010	0.0397	0.0396	0.0001
OLS	Sentinel-2	AOI-03; OLS (W2023)	18,857	0.011	0.000	0.011	0.0399	0.0398	0.0001
OLS	Sentinel-2	AOI-03; OLS (W2023_2024)	18,857	0.009	0.000	0.009	0.0397	0.0396	0.0000
OLS	Sentinel-2	AOI-03; OLS (W2024)	18,857	0.008	0.000	0.008	0.0397	0.0396	0.0000

Table S6: Clipping diagnostics for all DPM endpoint configurations and OLS histories: MAE and Bias.

Model	Sensor	AOI/configuration	n	MAE raw	MAE clipped	Bias raw	Bias clipped
DPM	Landsat 8/9	AOI-00; P1/P99	20,251	0.2185	0.2173	0.2044	0.2052
DPM	Landsat 8/9	AOI-00; P10/P90	20,251	0.2946	0.2079	-0.0493	0.0170
DPM	Landsat 8/9	AOI-00; P2/P98	20,251	0.2121	0.2076	0.1818	0.1851
DPM	Landsat 8/9	AOI-00; P5/P95	20,251	0.2068	0.1807	0.0853	0.1063
DPM	Landsat 8/9	AOI-01; P1/P99	14,738	0.4560	0.4497	0.4499	0.4497
DPM	Landsat 8/9	AOI-01; P10/P90	14,738	0.5486	0.4891	0.4987	0.4891
DPM	Landsat 8/9	AOI-01; P2/P98	14,738	0.4631	0.4519	0.4526	0.4519
DPM	Landsat 8/9	AOI-01; P5/P95	14,738	0.4935	0.4677	0.4709	0.4677
DPM	Landsat 8/9	AOI-02; P1/P99	7,600	0.1917	0.1911	0.1866	0.1867
DPM	Landsat 8/9	AOI-02; P10/P90	7,600	0.2418	0.1982	0.1020	0.1290
DPM	Landsat 8/9	AOI-02; P2/P98	7,600	0.1972	0.1957	0.1879	0.1883
DPM	Landsat 8/9	AOI-02; P5/P95	7,600	0.2086	0.1994	0.1736	0.1780
DPM	Landsat 8/9	AOI-03; P1/P99	14,402	0.1748	0.1740	0.1725	0.1722
DPM	Landsat 8/9	AOI-03; P10/P90	14,402	0.3501	0.3032	0.3291	0.2953
DPM	Landsat 8/9	AOI-03; P2/P98	14,402	0.1890	0.1868	0.1854	0.1842
DPM	Landsat 8/9	AOI-03; P5/P95	14,402	0.2422	0.2300	0.2337	0.2252
DPM	MODIS	AOI-00; P1/P99	20,256	0.1973	0.1958	0.1850	0.1861
DPM	MODIS	AOI-00; P10/P90	20,256	0.2759	0.2066	-0.0523	-0.0014
DPM	MODIS	AOI-00; P2/P98	20,256	0.1830	0.1778	0.1498	0.1538
DPM	MODIS	AOI-00; P5/P95	20,256	0.1906	0.1654	0.0513	0.0711
DPM	MODIS	AOI-01; P1/P99	19,878	0.3870	0.3816	0.3812	0.3816
DPM	MODIS	AOI-01; P10/P90	19,878	0.5091	0.4630	0.4778	0.4630
DPM	MODIS	AOI-01; P2/P98	19,878	0.4070	0.3977	0.3985	0.3977
DPM	MODIS	AOI-01; P5/P95	19,878	0.4527	0.4314	0.4367	0.4314
DPM	MODIS	AOI-02; P1/P99	16,568	0.1868	0.1863	0.1833	0.1833
DPM	MODIS	AOI-02; P10/P90	16,568	0.2278	0.1944	0.1272	0.1465
DPM	MODIS	AOI-02; P2/P98	16,568	0.1941	0.1928	0.1873	0.1875
DPM	MODIS	AOI-02; P5/P95	16,568	0.2021	0.1953	0.1767	0.1796
DPM	MODIS	AOI-03; P1/P99	17,860	0.1907	0.1900	0.1898	0.1895
DPM	MODIS	AOI-03; P10/P90	17,860	0.3719	0.3258	0.3518	0.3208
DPM	MODIS	AOI-03; P2/P98	17,860	0.2075	0.2051	0.2055	0.2041
DPM	MODIS	AOI-03; P5/P95	17,860	0.2646	0.2523	0.2585	0.2500
DPM	Sentinel-2	AOI-00; P1/P99	19,233	0.1781	0.1770	0.1550	0.1556
DPM	Sentinel-2	AOI-00; P10/P90	19,233	0.2801	0.2212	-0.0399	-0.0008
DPM	Sentinel-2	AOI-00; P2/P98	19,233	0.1745	0.1706	0.1273	0.1300
DPM	Sentinel-2	AOI-00; P5/P95	19,233	0.1975	0.1777	0.0446	0.0592
DPM	Sentinel-2	AOI-01; P1/P99	19,878	0.3915	0.3780	0.3806	0.3780
DPM	Sentinel-2	AOI-01; P10/P90	19,878	0.5392	0.4676	0.4900	0.4676
DPM	Sentinel-2	AOI-01; P2/P98	19,878	0.4247	0.4042	0.4091	0.4042
DPM	Sentinel-2	AOI-01; P5/P95	19,878	0.4783	0.4398	0.4511	0.4398
DPM	Sentinel-2	AOI-02; P1/P99	18,534	0.1564	0.1560	0.1430	0.1429
DPM	Sentinel-2	AOI-02; P10/P90	18,534	0.2164	0.1926	0.1295	0.1393
DPM	Sentinel-2	AOI-02; P2/P98	18,534	0.1661	0.1650	0.1483	0.1483
DPM	Sentinel-2	AOI-02; P5/P95	18,534	0.1865	0.1819	0.1566	0.1569
DPM	Sentinel-2	AOI-03; P1/P99	18,857	0.1730	0.1720	0.1699	0.1693
DPM	Sentinel-2	AOI-03; P10/P90	18,857	0.3496	0.3074	0.3291	0.2988
DPM	Sentinel-2	AOI-03; P2/P98	18,857	0.1958	0.1927	0.1913	0.1892
DPM	Sentinel-2	AOI-03; P5/P95	18,857	0.2600	0.2460	0.2508	0.2405
OLS	Landsat 8/9	AOI-00; OLS (W2022)	20,251	0.0390	0.0390	-0.0086	-0.0086
OLS	Landsat 8/9	AOI-00; OLS (W2022_2023)	20,251	0.0389	0.0389	-0.0060	-0.0060
OLS	Landsat 8/9	AOI-00; OLS (W2022_2024)	20,251	0.0398	0.0398	-0.0136	-0.0136
OLS	Landsat 8/9	AOI-00; OLS (W2023)	20,251	0.0389	0.0389	-0.0031	-0.0031
OLS	Landsat 8/9	AOI-00; OLS (W2023_2024)	20,251	0.0402	0.0402	-0.0152	-0.0152
OLS	Landsat 8/9	AOI-00; OLS (W2024)	20,251	0.0435	0.0435	-0.0266	-0.0266
OLS	Landsat 8/9	AOI-01; OLS (W2022)	14,738	0.0007	0.0005	-0.0003	-0.0000
OLS	Landsat 8/9	AOI-01; OLS (W2022_2023)	14,738	0.0008	0.0006	-0.0002	0.0000
OLS	Landsat 8/9	AOI-01; OLS (W2022_2024)	14,738	0.0008	0.0006	-0.0001	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2023)	14,738	0.0008	0.0006	-0.0001	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2023_2024)	14,738	0.0008	0.0006	-0.0001	0.0001
OLS	Landsat 8/9	AOI-01; OLS (W2024)	14,738	0.0008	0.0006	-0.0001	0.0001
OLS	Landsat 8/9	AOI-02; OLS (W2022)	7,600	0.0435	0.0435	0.0023	0.0023
OLS	Landsat 8/9	AOI-02; OLS (W2022_2023)	7,600	0.0439	0.0439	0.0091	0.0091
OLS	Landsat 8/9	AOI-02; OLS (W2022_2024)	7,600	0.0442	0.0442	0.0107	0.0107
OLS	Landsat 8/9	AOI-02; OLS (W2023)	7,600	0.0484	0.0484	0.0254	0.0254
OLS	Landsat 8/9	AOI-02; OLS (W2023_2024)	7,600	0.0484	0.0484	0.0256	0.0256
OLS	Landsat 8/9	AOI-02; OLS (W2024)	7,600	0.0449	0.0449	0.0143	0.0143
OLS	Landsat 8/9	AOI-03; OLS (W2022)	14,402	0.0255	0.0254	-0.0001	0.0001
OLS	Landsat 8/9	AOI-03; OLS (W2022_2023)	14,402	0.0276	0.0275	0.0089	0.0090
OLS	Landsat 8/9	AOI-03; OLS (W2022_2024)	14,402	0.0273	0.0272	0.0081	0.0082
OLS	Landsat 8/9	AOI-03; OLS (W2023)	14,402	0.0337	0.0336	0.0220	0.0221
OLS	Landsat 8/9	AOI-03; OLS (W2023_2024)	14,402	0.0291	0.0290	0.0130	0.0130
OLS	Landsat 8/9	AOI-03; OLS (W2024)	14,402	0.0270	0.0269	0.0069	0.0070
OLS	MODIS	AOI-00; OLS (W2022)	20,256	0.0309	0.0309	-0.0014	-0.0014
OLS	MODIS	AOI-00; OLS (W2022_2023)	20,256	0.0310	0.0310	-0.0006	-0.0006
OLS	MODIS	AOI-00; OLS (W2022_2024)	20,256	0.0312	0.0312	-0.0066	-0.0066
OLS	MODIS	AOI-00; OLS (W2023)	20,256	0.0311	0.0311	0.0006	0.0006
OLS	MODIS	AOI-00; OLS (W2023_2024)	20,256	0.0315	0.0315	-0.0086	-0.0086
OLS	MODIS	AOI-00; OLS (W2024)	20,256	0.0337	0.0337	-0.0179	-0.0179
OLS	MODIS	AOI-01; OLS (W2022)	19,878	0.0007	0.0006	-0.0000	0.0002
OLS	MODIS	AOI-01; OLS (W2022_2023)	19,878	0.0007	0.0005	-0.0001	0.0001
OLS	MODIS	AOI-01; OLS (W2022_2024)	19,878	0.0007	0.0005	-0.0001	0.0000
OLS	MODIS	AOI-01; OLS (W2023)	19,878	0.0007	0.0005	-0.0001	0.0001
OLS	MODIS	AOI-01; OLS (W2023_2024)	19,878	0.0006	0.0005	-0.0001	-0.0000
OLS	MODIS	AOI-01; OLS (W2024)	19,878	0.0006	0.0005	-0.0002	-0.0001
OLS	MODIS	AOI-02; OLS (W2022)	16,568	0.0425	0.0425	0.0254	0.0255
OLS	MODIS	AOI-02; OLS (W2022_2023)	16,568	0.0397	0.0397	0.0181	0.0181
OLS	MODIS	AOI-02; OLS (W2022_2024)	16,568	0.0408	0.0408	0.0213	0.0213
OLS	MODIS	AOI-02; OLS (W2023)	16,568	0.0370	0.0370	0.0029	0.0029
OLS	MODIS	AOI-02; OLS (W2023_2024)	16,568	0.0398	0.0398	0.0185	0.0185
OLS	MODIS	AOI-02; OLS (W2024)	16,568	0.0430	0.0430	0.0264	0.0264
OLS	MODIS	AOI-03; OLS (W2022)	17,860	0.0235	0.0234	0.0111	0.0112
OLS	MODIS	AOI-03; OLS (W2022_2023)	17,860	0.0253	0.0252	0.0157	0.0158
OLS	MODIS	AOI-03; OLS (W2022_2024)	17,860	0.0228	0.0226	0.0084	0.0085
OLS	MODIS	AOI-03; OLS (W2023)	17,860	0.0276	0.0276	0.0200	0.0201
OLS	MODIS	AOI-03; OLS (W2023_2024)	17,860	0.0224	0.0223	0.0072	0.0074
OLS	MODIS	AOI-03; OLS (W2024)	17,860	0.0207	0.0204	-0.0046	-0.0043
OLS	Sentinel-2	AOI-00; OLS (W2022)	19,233	0.0512	0.0512	-0.0249	-0.0249
OLS	Sentinel-2	AOI-00; OLS (W2022_2023)	19,233	0.0511	0.0511	-0.0248	-0.0248
OLS	Sentinel-2	AOI-00; OLS (W2022_2024)	19,233	0.0516	0.0516	-0.0261	-0.0261
OLS	Sentinel-2	AOI-00; OLS (W2023)	19,233	0.0513	0.0513	-0.0253	-0.0253
OLS	Sentinel-2	AOI-00; OLS (W2023_2024)	19,233	0.0517	0.0517	-0.0265	-0.0265
OLS	Sentinel-2	AOI-00; OLS (W2024)	19,233	0.0537	0.0537	-0.0302	-0.0302
OLS	Sentinel-2	AOI-01; OLS (W2022)	19,878	0.0007	0.0004	-0.0005	-0.0002
OLS	Sentinel-2	AOI-01; OLS (W2022_2023)	19,878	0.0007	0.0003	-0.0006	-0.0002
OLS	Sentinel-2	AOI-01; OLS (W2022_2024)	19,878	0.0007	0.0003	-0.0006	-0.0002
OLS	Sentinel-2	AOI-01; OLS (W2023)	19,878	0.0008	0.0003	-0.0007	-0.0002
OLS	Sentinel-2	AOI-01; OLS (W2023_2024)	19,878	0.0008	0.0003	-0.0007	-0.0002
OLS	Sentinel-2	AOI-01; OLS (W2024)	19,878	0.0007	0.0003	-0.0006	-0.0002
OLS	Sentinel-2	AOI-02; OLS (W2022)	18,534	0.0467	0.0467	-0.0160	-0.0160
OLS	Sentinel-2	AOI-02; OLS (W2022_2023)	18,534	0.0453	0.0453	-0.0071	-0.0071
OLS	Sentinel-2	AOI-02; OLS (W2022_2024)	18,534	0.0450	0.0450	-0.0025	-0.0025

Model	Sensor	AOI/configuration	n	MAE raw	MAE clipped	Bias raw	Bias clipped
OLS	Sentinel-2	AOI-02; OLS (W2023)	18,534	0.0470	0.0470	0.0156	0.0156
OLS	Sentinel-2	AOI-02; OLS (W2023_2024)	18,534	0.0456	0.0456	0.0079	0.0079
OLS	Sentinel-2	AOI-02; OLS (W2024)	18,534	0.0453	0.0453	0.0053	0.0053
OLS	Sentinel-2	AOI-03; OLS (W2022)	18,857	0.0275	0.0274	-0.0022	-0.0021
OLS	Sentinel-2	AOI-03; OLS (W2022_2023)	18,857	0.0273	0.0272	-0.0031	-0.0030
OLS	Sentinel-2	AOI-03; OLS (W2022_2024)	18,857	0.0276	0.0275	-0.0019	-0.0018
OLS	Sentinel-2	AOI-03; OLS (W2023)	18,857	0.0272	0.0271	-0.0041	-0.0040
OLS	Sentinel-2	AOI-03; OLS (W2023_2024)	18,857	0.0276	0.0276	-0.0017	-0.0016
OLS	Sentinel-2	AOI-03; OLS (W2024)	18,857	0.0281	0.0281	0.0004	0.0005

Table S7: Supplementary Table S7. Aggregation-order sensitivity for preferred 2025 Multi-AOI histories. Route B is NDVI from mean matched-support RED and NIR; Δ RMSE is Route B minus Route A.

Sensor	AOI	Δ RMSE	Mean $ \Delta$ NDVI	Max $ \Delta$ NDVI	Retention	History	RO direction
Landsat	AOI-00	-0.001458	0.004409	0.115172	1.000	no	no
Landsat	AOI-01	+0.000018	0.000238	0.033406	1.000	no	no
Landsat	AOI-02	+0.000493	0.004539	0.110796	1.000	no	no
Landsat	AOI-03	-0.000710	0.003148	0.061446	1.000	no	no
MODIS	AOI-00	-0.000695	0.001523	0.199975	1.000	no	no
MODIS	AOI-01	+0.000008	0.000101	0.101627	1.000	no	no
MODIS	AOI-02	+0.000480	0.001790	0.072271	1.000	no	no
MODIS	AOI-03	-0.000005	0.001356	0.046882	1.000	yes	no
Sentinel-2	AOI-00	-0.000508	0.006719	0.421059	1.000	no	yes
Sentinel-2	AOI-01	+0.000072	0.000270	0.151513	1.000	no	no
Sentinel-2	AOI-02	+0.000454	0.006864	0.176911	1.000	no	no
Sentinel-2	AOI-03	-0.000088	0.003566	0.192119	1.000	no	yes

Table S8: Supplementary Table S8. Non-overlapping temporal-composition sensitivity for preferred 2025 Multi-AOI histories. Operational and matched Δ RMSE are sensitivity minus primary.

Sensor	AOI	Retention	Operational Δ RMSE	Matched Δ RMSE	History	RO direction
Landsat	AOI-00	0.988445	+0.002418	+0.002503	yes	no
Landsat	AOI-01	0.651445	-0.000146	+0.000024	no	no
Landsat	AOI-02	0.228289	+0.001467	+0.001884	yes	yes
Landsat	AOI-03	0.647271	+0.001526	+0.000837	no	no
MODIS	AOI-00	0.920271	+0.003063	+0.002771	yes	no
MODIS	AOI-01	0.991750	+0.000006	+0.000021	no	no
MODIS	AOI-02	0.132183	+0.008773	+0.017763	yes	no
MODIS	AOI-03	0.703359	+0.004847	+0.006436	yes	yes
Sentinel-2	AOI-00	0.665991	+0.009703	+0.007073	no	no
Sentinel-2	AOI-01	0.976406	+0.000011	+0.000071	yes	no
Sentinel-2	AOI-02	0.681828	+0.001203	+0.004428	no	no
Sentinel-2	AOI-03	0.561595	+0.002327	+0.004515	yes	yes

Table S9: Supplementary Table S9. Landsat aerosol-QA sensitivity for preferred 2025 Multi-AOI histories. Each mode is additional to primary Collection 2 Level-2 surface-reflectance processing; Δ RMSE is sensitivity minus primary.

Mode	AOI	Retention	Operational Δ RMSE	Matched Δ RMSE	History	RO direction
Exclude high	AOI-00	0.999111	-0.001201	-0.001208	no	no
Exclude high	AOI-01	0.698942	-0.000061	+0.000111	no	yes
Exclude high	AOI-02	0.668816	+0.010846	+0.014958	yes	no
Exclude high	AOI-03	0.971809	-0.000275	+0.000183	no	yes
Strict	AOI-00	0.995408	+0.006481	+0.006506	no	yes
Strict	AOI-01	0.580812	+0.000115	+0.000306	yes	yes
Strict	AOI-02	0.395526	+0.027036	+0.033996	yes	yes
Strict	AOI-03	0.811901	+0.002709	+0.005027	no	yes
Valid retrieval/no high	AOI-00	0.998469	+0.000585	+0.000575	no	no
Valid retrieval/no high	AOI-01	0.644457	-0.000105	+0.000080	no	yes
Valid retrieval/no high	AOI-02	0.603158	+0.022310	+0.028278	yes	yes
Valid retrieval/no high	AOI-03	0.917720	+0.000608	+0.001459	no	yes